

Calcium imaging report

helobdella_LeechNo2_elavgcamp6m-17

Pan-neuronal GCaMP6m, 10x juvenile leech ganglion, after dopamine

Condensed report. A single strong, clock-like, ganglion-wide calcium oscillation at 0.37 Hz (period 2.7 s), sustained for the full ~240 s recording and spatially synchronous, with no traveling wave.

What we did

- Five-stage pipeline on the raw movie (692 frames, 700x700, fs 2.882 Hz, ~240 s, single-channel pan-neuronal GCaMP6m, 10x juvenile leech, after dopamine).
- Stage 1 QC and motion: photobleaching, rigid and non-rigid drift, correlation image.
- Stage 2 segmentation: suite2p (24 ROIs, kept as a cross-check) and a primary correlation-blob plus watershed method (180 active patches).
- Stage 3 and 4: temporal (spectra, spectrogram, autocorrelation, Hilbert) and spatial (per-pixel power and phase, traveling-wave test, region coherence) characterisation.
- Intricate visualization pass: 25 detailed figures plus a cycle-averaged movie.
- 10x and a preliminary line, so everything is read at the ganglion or regional level, not as single neurons. The 180 patches are active regions, not confirmed cells.

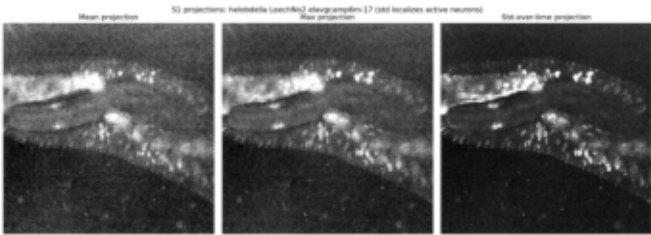
Headline result and how to interpret it

- One strong, clock-like, ganglion-wide calcium oscillation at 0.372 Hz (period 2.69 s), sustained undamped for the full ~240 s and synchronous across the ganglion.
- Frequency: dominant peak 0.372 Hz with a 0.73 Hz harmonic, ~230x above the spectral floor, well below the 1.44 Hz Nyquist, so the fundamental is not aliased.
- Regularity: 88 cycles, inter-peak interval CV 0.077, tight Poincare cluster, stable spectrogram band.
- Synchrony: vertical kymograph stripes, whole-ganglion brightening together, flat phase-versus-position, inter-region coherence ~0.88. No traveling wave.
- Nuance: a strong subset carries the rhythm. ~93% of patches peak within 0.05 Hz of 0.37 Hz and ~40% follow the ganglion mean at $r > 0.5$; the rest are noisy (low-photon 10x).
- Interpretation: a synchronous ganglion-wide oscillation suggests a shared network drive. Labelled after dopamine, but there is no pre-dopamine baseline here, so no causal claim. Motion correction was verified unnecessary (0 px drift, ~0.5% bleaching).

Key numbers

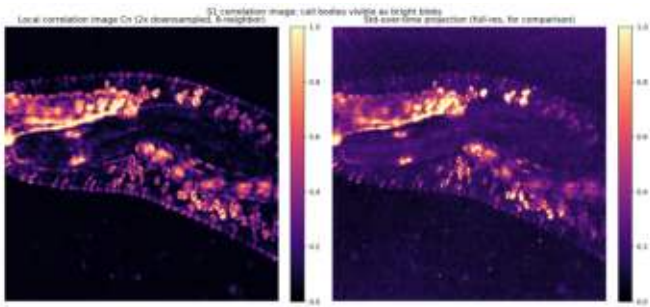
frame rate	2.882 Hz
duration	~240 s (692 frames)
dominant rhythm	0.372 Hz (period 2.69 s)
cycles / interval CV	88 / 0.077
active patches (primary)	180 (suite2p cross-check 24)
inter-region coherence at 0.37 Hz	~0.88
tissue pixels rhythmic	~93%
phase circular std (tissue)	1.13 rad
photobleaching / rigid drift	~0.5% / 0 px (no motion correction)

Stage 1: quality control and motion



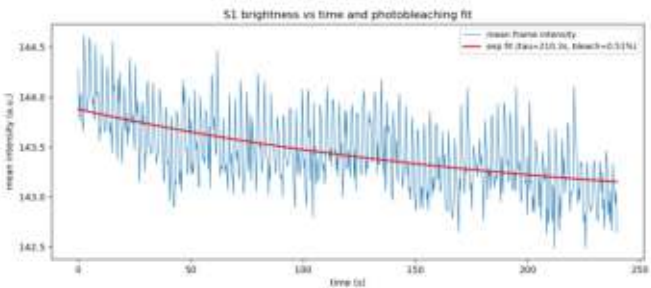
s1_projections.png

three summary images of the movie collapsed over time: the mean, the



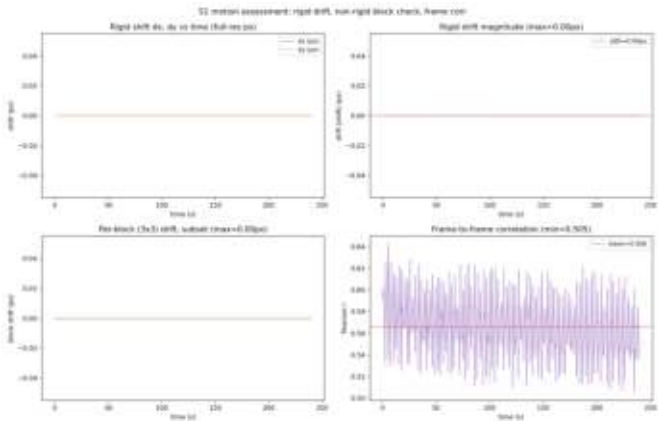
s1_correlation_image.png

the local correlation image (Cn), the cleanest map of cell bodies, beside



s1_brightness_bleach.png

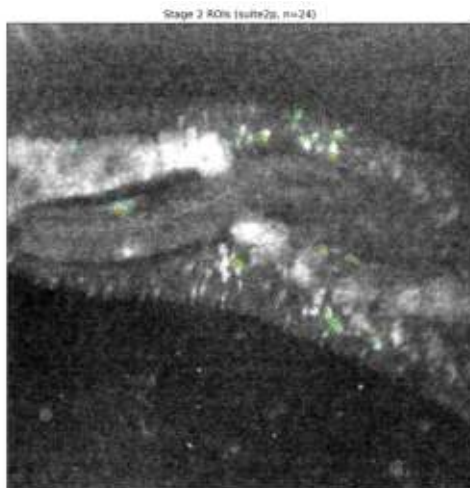
photobleaching check



s1_motion.png

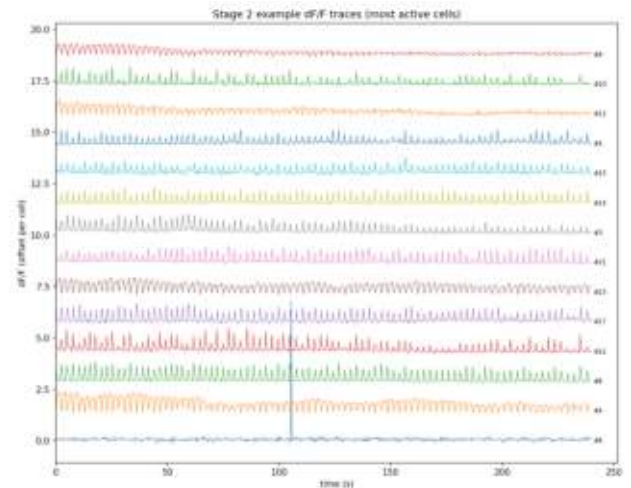
the motion-correction decision evidence

Stage 2 and 2b: segmentation and signal extraction



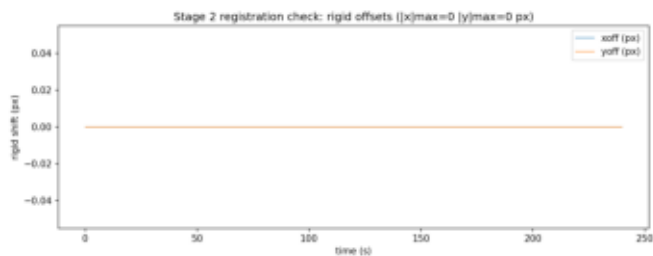
s2_rois_on_mean.png

suite2p's detected ROIs (n = 24) on the mean image



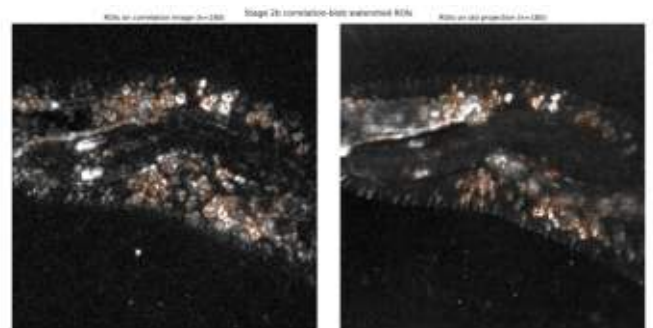
s2_example_traces.png

dF/F traces of the most active suite2p ROIs, stacked with vertical offsets



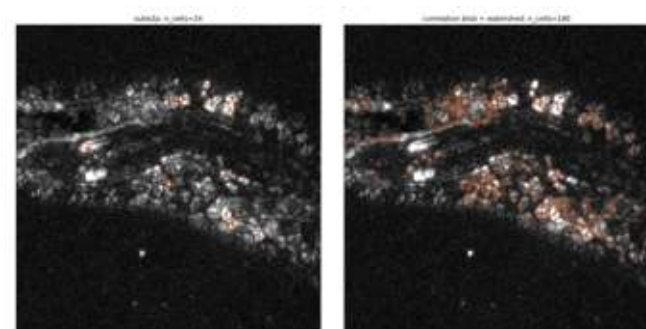
s2_registration_check.png

confirmation that registration was a no-op



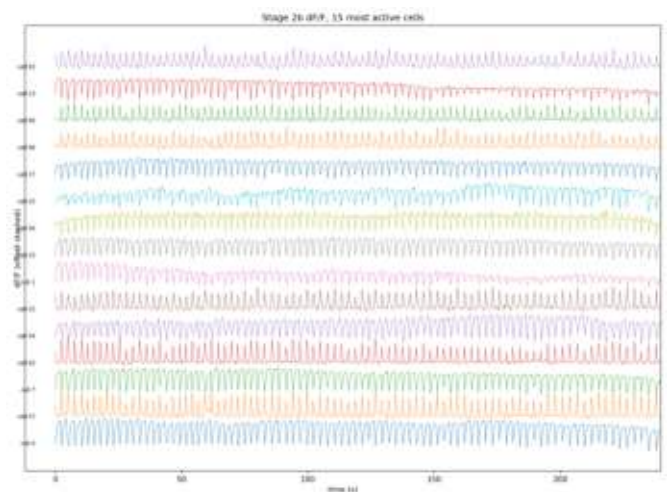
s2b_rois_on_corr.png

the primary segmentation, 180 ROIs from correlation-blob + watershed



s2b_method_comparison.png

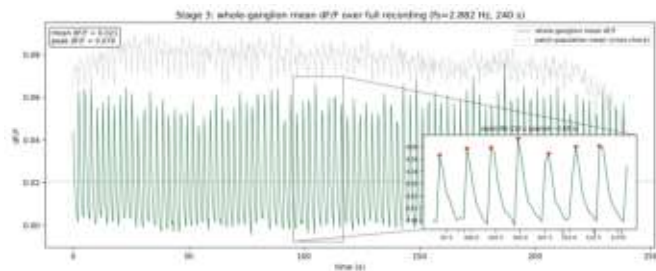
suite2p (24) vs correlation-blob (180) side by side on the same image



s2b_example_traces.png

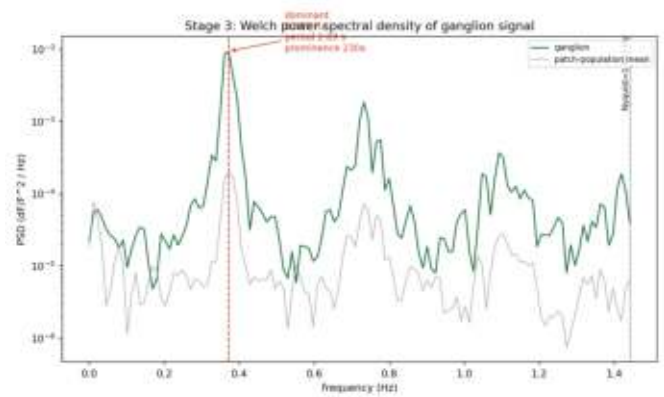
dF/F of the 15 most active patches from the primary method

Stage 3: temporal structure of the rhythm



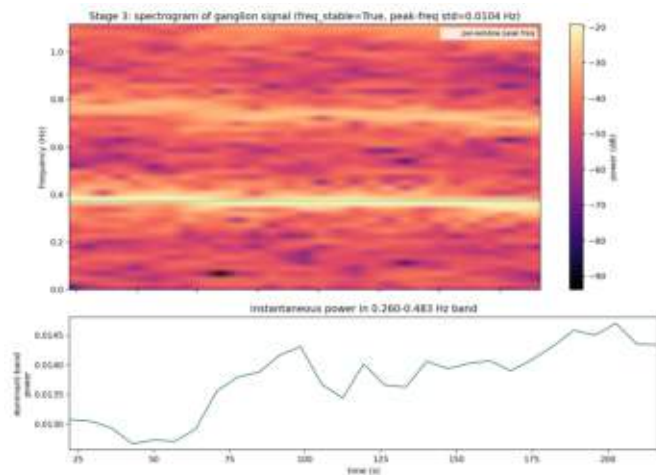
s3_ganglion_signal.png

the whole-ganglion mean dF/F over the full recording, with a 20 s zoom inset



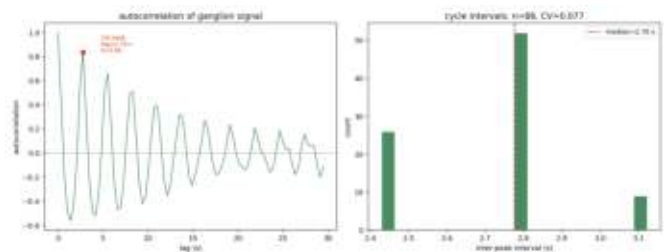
s3_spectrum.png

Welch power spectral density of the ganglion signal (and patch-population)



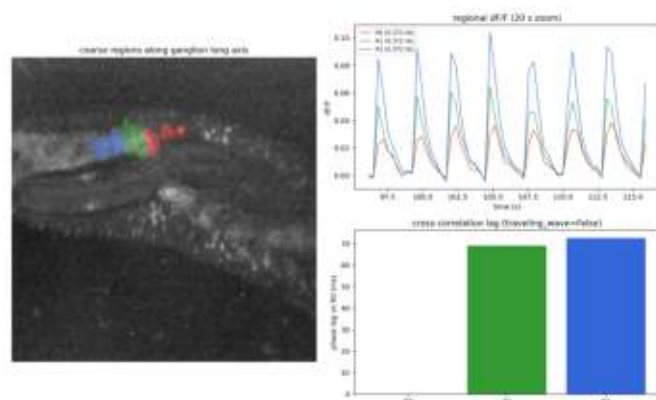
s3_spectrogram.png

how the frequency content evolves over time



s3_autocorr_regularity.png

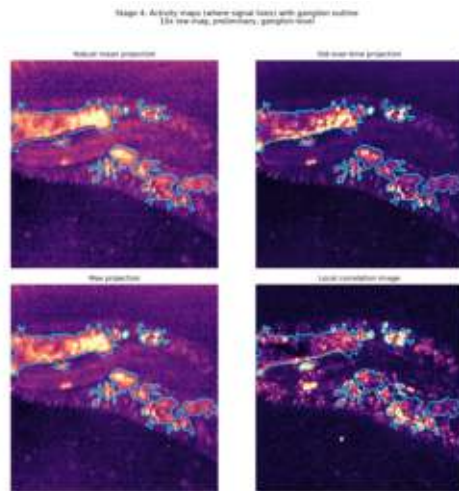
how clock-like the rhythm is



s3_regional_comparison.png

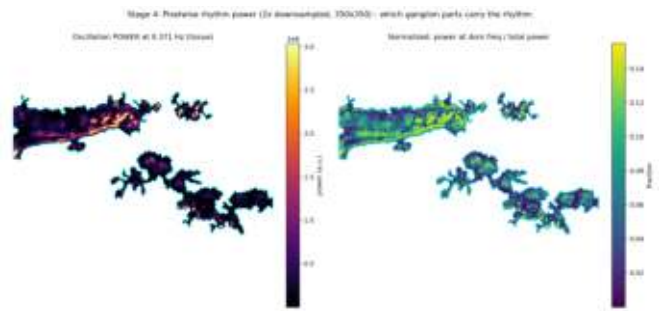
do coarse parts of the ganglion share the rhythm and its timing

Stage 4: spatial structure of the rhythm



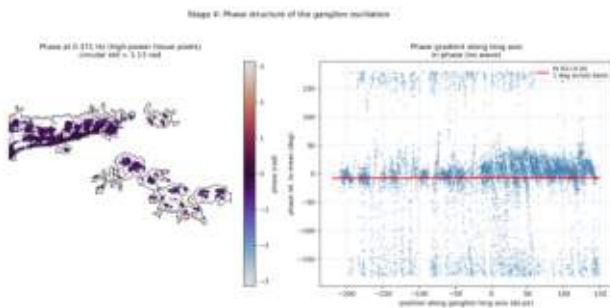
s4_activity_maps.png

where signal lives, as a panel of mean / std / max / correlation projections



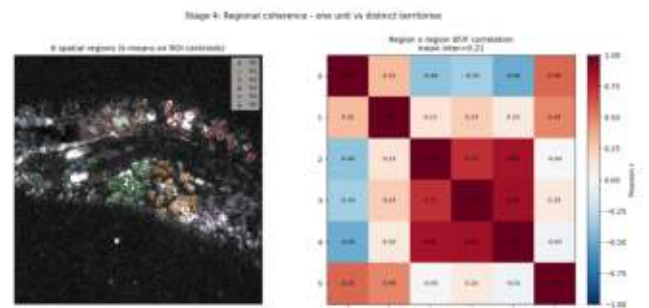
s4_pixelwise_rhythm_power.png

which pixels carry the 0.37 Hz rhythm



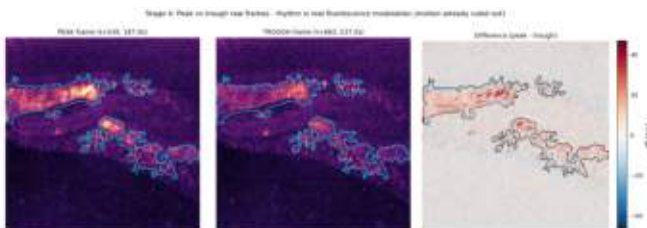
s4_phase_map.png

the timing (phase) of the oscillation across space



s4_regional_correlation.png

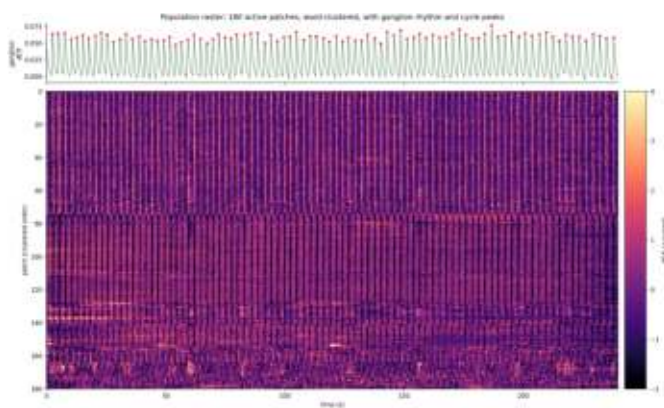
are there distinct functional territories



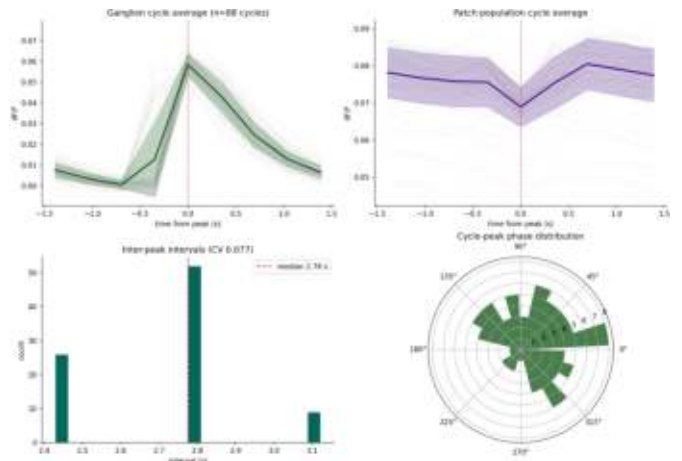
s4_peak_trough_frames.png

a sanity check that the rhythm is real fluorescence, not noise

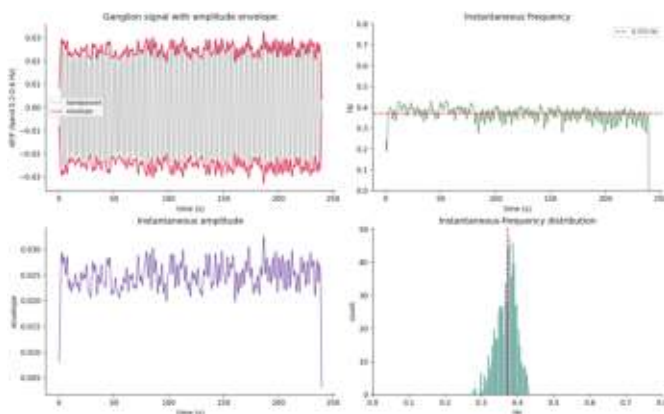
Intricate set: population / temporal (fig_pop_*)



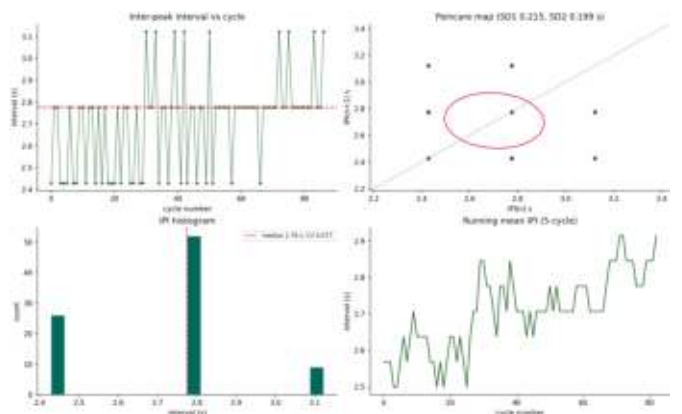
fig_pop_raster.png
the whole population at a glance



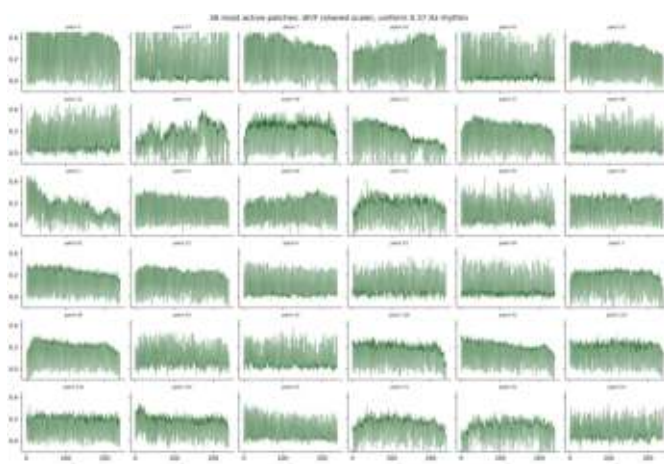
fig_pop_cycle_average.png
the average shape of one oscillation cycle



fig_pop_instantaneous.png
instantaneous amplitude and frequency via the Hilbert transform

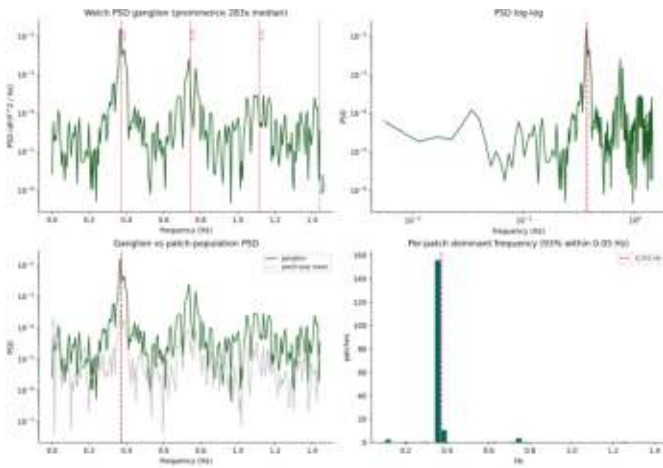


fig_pop_interval_dynamics.png
cycle-to-cycle timing regularity



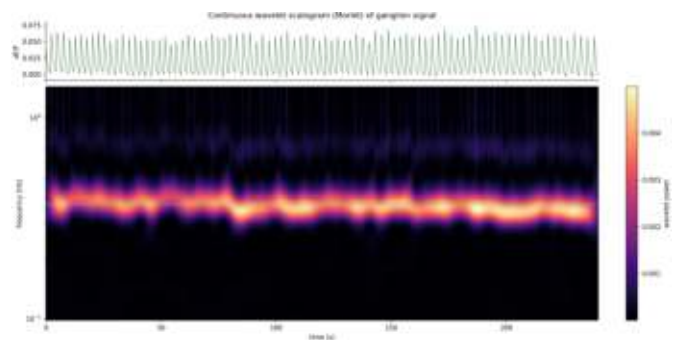
fig_pop_patch_overview.png
small-multiples of the 36 most active patches

Intricate set: spectral (fig_spec_*)



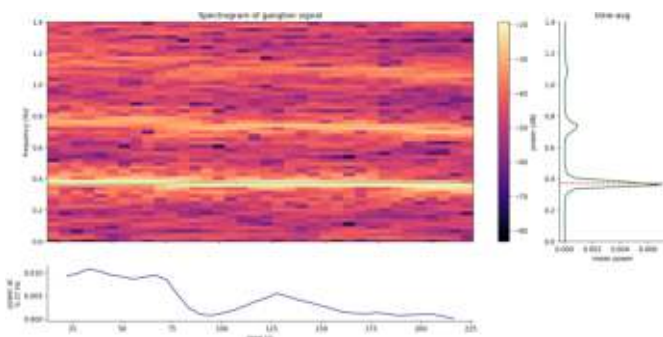
fig_spec_psd_detailed.png

a detailed look at the power spectrum



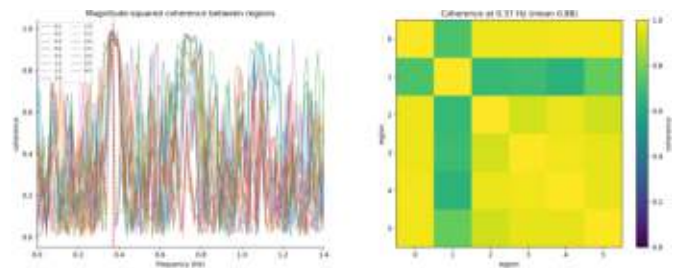
fig_spec_wavelet.png

a time-frequency (wavelet) view of the ganglion signal



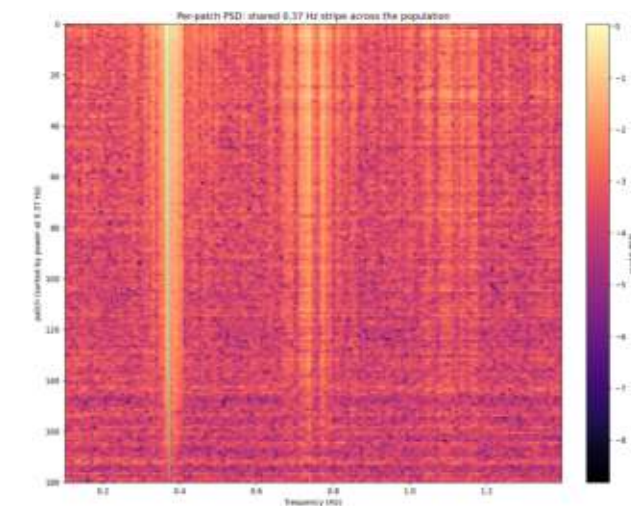
fig_spec_spectrogram_detailed.png

a high-resolution spectrogram with marginal summaries



fig_spec_coherence.png

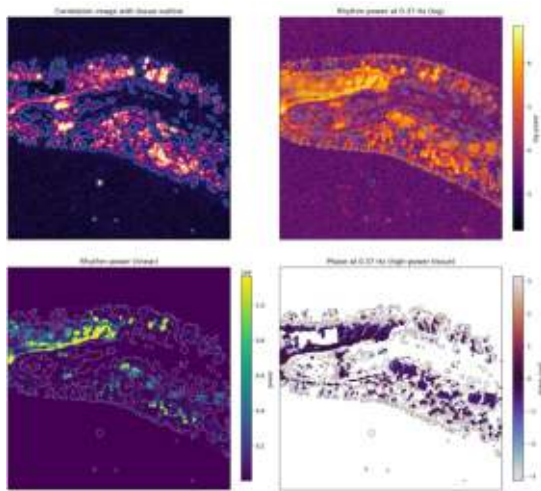
how strongly separated regions oscillate together, frequency by frequency



fig_spec_patch_spectra.png

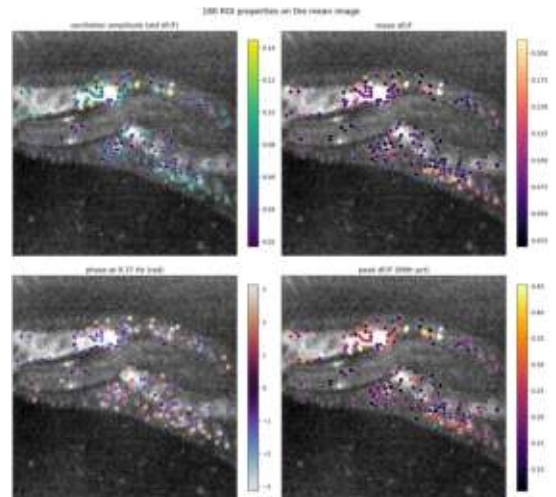
every patch's spectrum stacked into one image

Intricate set: spatial (fig_spatial_*)



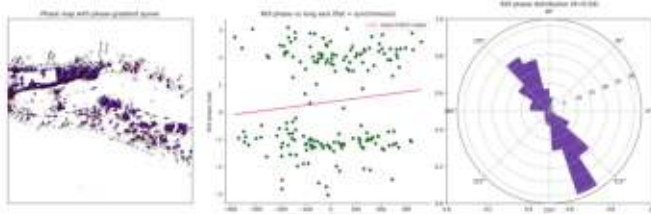
fig_spatial_power_phase.png

a 2x2 of correlation image, rhythm power (log and linear), and phase



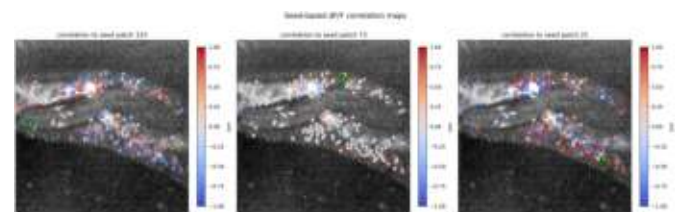
fig_spatial_roi_properties.png

the 180 ROIs colored by four properties on the anatomy



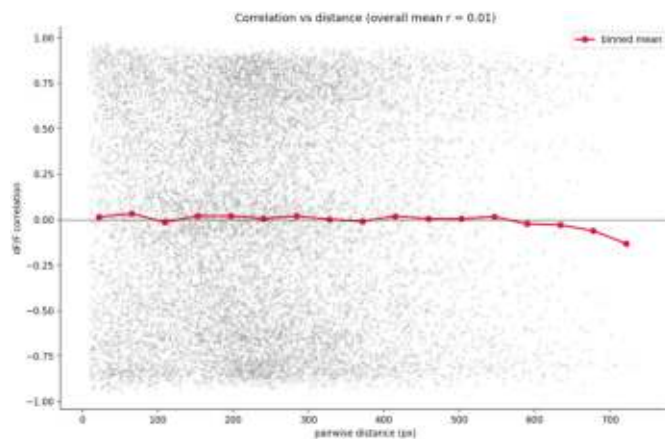
fig_spatial_phase_gradient.png

an explicit traveling-wave test



fig_spatial_seed_correlation.png

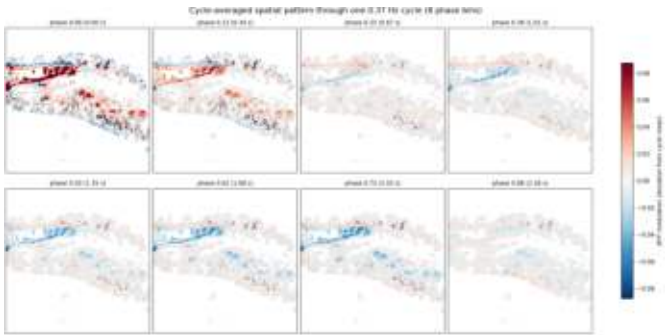
how far co-activity reaches from a chosen patch



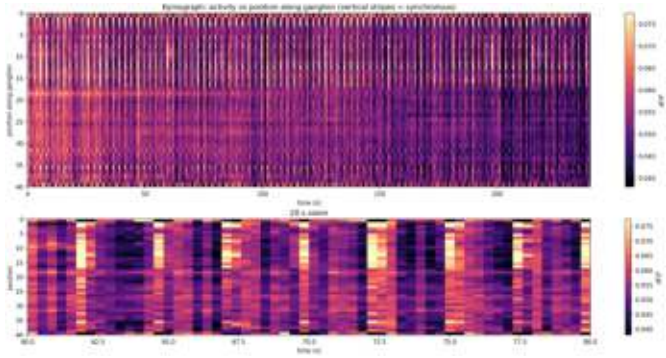
fig_spatial_distance_correlation.png

does proximity predict co-activity

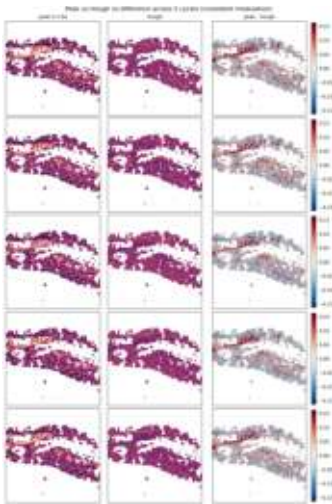
Intricate set: cycle-resolved spatiotemporal (fig_cycle_*)



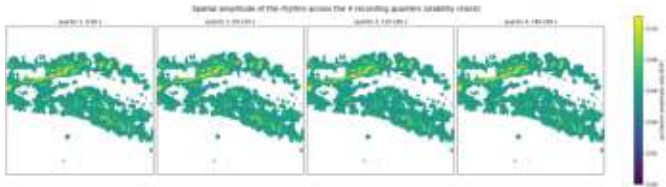
fig_cycle_phase_montage.png
one average oscillation cycle played out in space



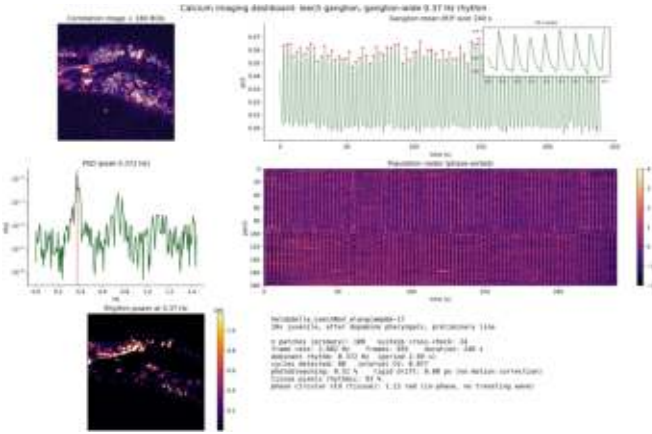
fig_cycle_kymograph.png
space (along the ganglion) vs time, the clearest wave test



fig_cycle_peak_trough_gallery.png
cycle-to-cycle consistency of the spatial pattern

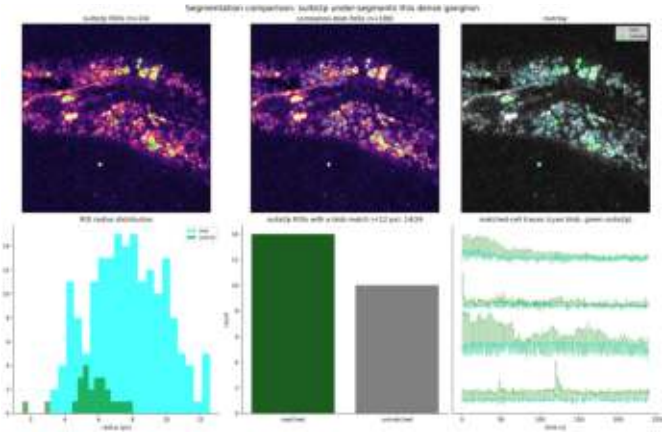


fig_cycle_amplitude_map_over_time.png
is the rhythm's spatial amplitude stable over the recording



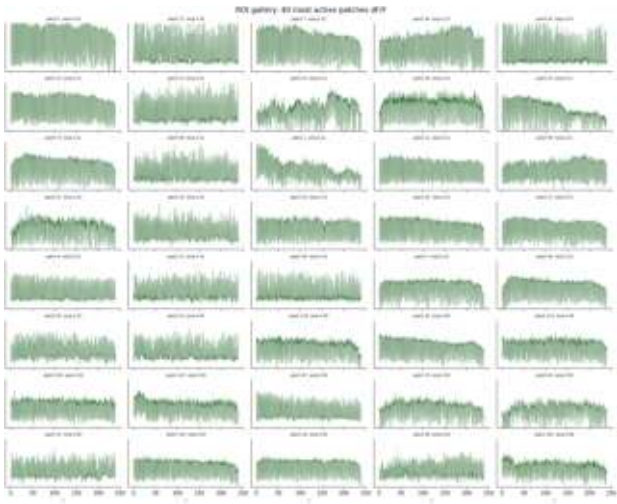
fig_dash_master.png

the one-page summary of the whole analysis



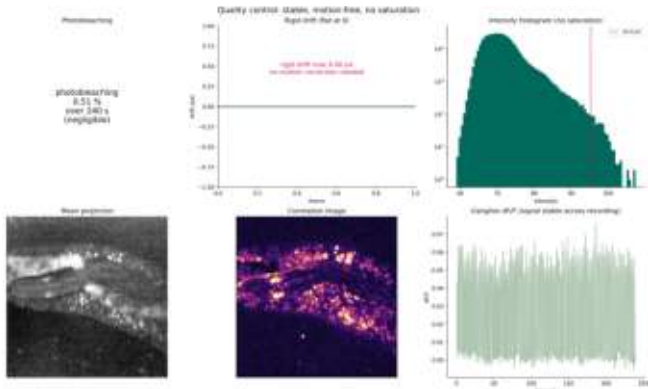
fig_dash_segmentation_compare.png

a detailed suite2p vs correlation-blob comparison



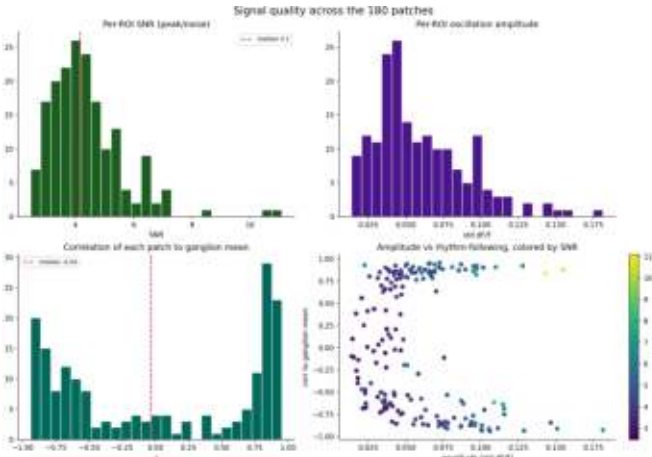
fig_dash_roi_gallery.png

a contact sheet of the 40 most active patches



fig_dash_qc_panel.png

a polished recap of the quality control



fig_dash_trace_quality.png

how good the per-patch signals are